Scattering-Attention Semi-Supervised-Guided Reinforced Region Proposal Network for Zero-Shot Detection on SAR Images

Dunyun He, Yucong He, and Gui Gao, *Senior Member, IEEE*

***Abstract*—Zero-shot detection (ZSD), which extends conventional detection models to recognize objects from previously unseen categories, represents one of the most promising strategies for reducing data annotation costs while enhancing model generalization capabilities. Generic ZSD methods mainly focus on excavating a strict mapping-transfer strategy, ignoring dataset deficiency and the heterogeneity between the visual and semantic domains in SAR imagery. This results in increased scattering noise interference and weak responses from proposal networks when dealing with unseen categories. In this article, we proposed a scattering-attention semi-supervised-guided reinforced region proposal (SSRP) network, a detection framework that first brings the zero-shot mechanism into the realm of SAR imagery. First, to address the issue of weak cross-domain feature response, the proposal optimization module (POM) is constructed to improve the ability to capture visual information and the discrimination of the unseen classes via a teacher–student model guided by soft labels. Second, to solve the problem of scattering noise interference, the scatter excitation learning module (SELM) is proposed to reduce background noise interference and focus on the local scattering center features of the foreground examples during training to achieve a tighter cross-domain feature mapping. Then, to overcome the challenge of dataset deficiency, the picture-to-text method $\mathcal{P}_2T$ is proposed to automatically generate descriptive terms for key attributes of SAR images, alleviating the issue of limited data availability, which is considered the main bottleneck in SAR-ZSD. We also improved the existing dataset to verify the effectiveness of the proposed method and push the development of SAR-ZSD further forward. Experimental results demonstrate that the SSRP outperforms the state-of-the-art generic methods and can improve 60% in recall and 19.59% in mAP.**

***Index Terms*—Semi-supervised, synthetic aperture radar (SAR), target detection, zero-shot detection (ZSD).**

I. Introduction

REMOTE sensing target detection technology based on synthetic aperture radar (SAR) imagery has been widely adopted in both military and civilian applications, owing to its unique advantages of all-weather, around-the-clock continuous observation [1]. In recent years, a proliferation of novel detection models based on deep convolutional neural networks (CNNs) has emerged, leading to substantial improvements in detection performance. Conventional object detection frameworks predominantly utilize a supervised learning approach using region proposals [2], where candidate regions likely to contain objects are generated, followed by simultaneous bounding box regression and classification for each proposal. However, the performance of these frameworks is critically dependent on large volumes of annotated training data and comprehensive labels for each class. In real-world scenarios, it is practically infeasible to thoroughly annotate all potential data samples, given that numerous novel or fine-grained object categories are constantly being identified. The annotation process for SAR imagery, in particular, not only requires specialized domain expertise but also entails considerable human effort. Moreover, generic methods [3], [4], [5], [6], [7] are inherently limited to recognizing only the classes seen during training, which significantly constrains their performance when faced with unseen categories. Unlike traditional detection methods that require prior training on all target classes, zero-shot detection (ZSD) leverages semantic attributes or textual descriptions associated with object classes to generalize to novel categories. This approach is widely regarded as one of the most promising strategies to alleviate data scarcity and address the inherent challenges posed by unseen classes. Recent advancements [8], [9], [10], [11], [12] highlight the critical need for developing techniques capable of accurately understanding and recognizing unseen targets. ZSD aims to unify the recognition and localization of novel objects within a single framework. Specifically, ZSD adheres to the conventional deep feature processing pipeline, where visual features and semantic vectors of seen class samples are projected into a shared embedding space. Within this shared space, a mapping function is employed to measure distances and establish virtual links between similar visual–semantic attribute features as closely as possible. The distinct characteristic of ZSD lies in the test phase, wherein the learned network is utilized to project the visual features and semantic representations of unseen classes into the shared space. Subsequently, search-ranking algorithms are employed to predict the corresponding labels for the unseen classes.

Received 18 November 2024; revised 22 April 2025, 3 September 2025, and 11 October 2025; accepted 6 November 2025. Date of publication 17 November 2025; date of current version 8 January 2026. This work was supported by the National Key Research and Development Program of China under Grant 2023YFB3905504. *(Dunyun He and Yucong He contributed equally to this work.) (Corresponding author: Gui Gao.)*

The authors are with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: dellar@126.com).

Digital Object Identifier 10.1109/TGRS.2025.3631095

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

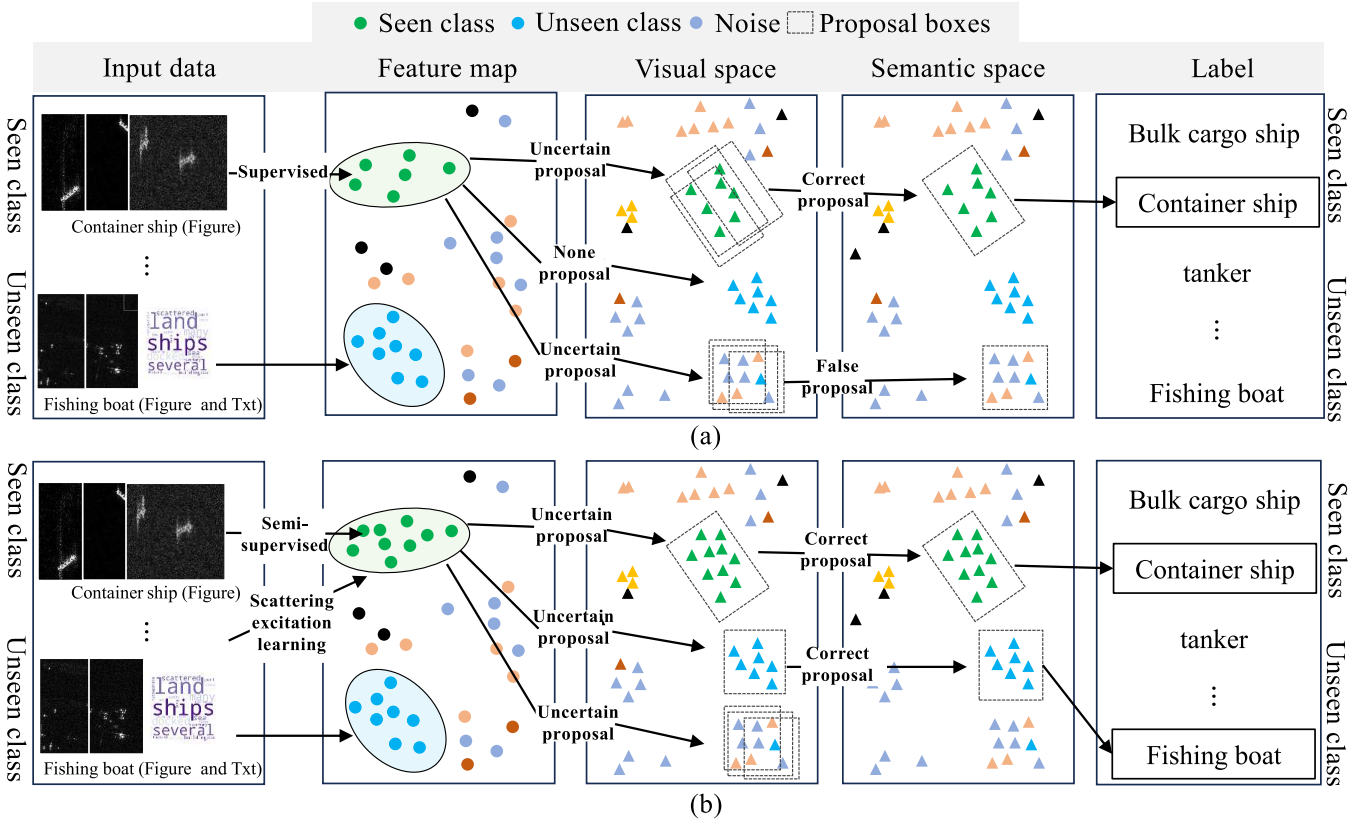


Fig. 1. Problems of SAR-ZSD. (a) Traditional ZSD methods extract limited usable information and often fail to generate proposal boxes for unseen classes during the visual space stage. (b) By introducing semi-supervised learning and scatter mechanism learning, we successfully enhanced attention to key features and addressed the issue of nonresponse to unseen domain migration.

A. Problem and Motivation

The application of ZSD to the SAR domain offers promising prospects for alleviating the substantial costs associated with manual annotation and improving the generalization capacity of detection models. However, the practical implementation of ZSD in SAR encounters certain challenges stemming from the intrinsic characteristics of SAR data, such as its complex imaging mechanisms and limited semantic interpretability. We conclude the problems of SAR-ZSD, as shown in Fig. 1.

1) In Terms of the Scarcity of SAR-ZSD Datasets: One of the main problems of applying ZSD to SAR is the lack of comprehensive datasets that are suitable for ZSD. Currently, ZSD primarily focuses on optical natural image data (e.g., face recognition and object detection), which benefits from vast amounts of data and relatively rich semantic labels. In contrast, the annotation process for SAR imagery heavily depends on domain experts, especially for ZSD tasks, which require more detailed textual descriptions of the images. Although publicly accessible datasets like SSDD [13] and OpenSARShip [14] exist, they are not readily applicable to ZSD. Moreover, the absence of standardized data formats and annotation protocols across different acquisition systems further hinders the improvement and integration of related datasets. Finally, due to the lack of a mature task definition and evaluation framework for SAR ZSD, current research remains in an exploratory phase, leading to uncertainty in dataset construction and dissemination.

2) In Terms of Weak Cross-Domain Feature Response to Unseen Classes: Another challenge in SAR-ZSD lies in the unique characteristics of SAR images, which complicate the transfer of proposal boxes from the seen to the unseen domain. Compared to optical images, SAR samples from the same class exhibit lower intraclass similarity, and the distinctions between different classes tend to be subtle and challenging to discern. This increases the complexity of designing mechanisms for attribute sharing and semantic transfer across intraclass and interclass features in ZSD tasks. Generic embedding-based methods have attempted to utilize feature similarities between seen classes and unseen data to improve the alignment between optical semantic and visual spaces. However, challenges in defining SAR image categories and attributes, as well as difficulties in constructing prior knowledge, can lead to serious issues such as the propagation of incorrect pseudo-labels and a reduction in semantic information. Therefore, the idea of self-supervised learning should be applied to ZSD to expand the model's feature perception and reduce missed detections during the proposal stage.

3) In Terms of Scattering Noise Interference in SAR Images: The sensitivity of SAR images to material properties and the complex interactions with different surfaces, leading to shadow effects and speckle noise [15], greatly complicates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

the description of target categories in ZSD tasks. Scattering noise substantially impairs the extraction of visual features, leading to misalignment in the visual–semantic embedding space and diminishing the model's capacity to infer unseen classes. Moreover, the presence of noise further complicates the generation of virtual samples, undermining the effectiveness of generative models in ZSD. Generic methods typically mitigate noise interference by integrating multiscale feature interactions [16]. However, due to the redundancy in features across different scales, attention to important features may be weakened [17]. In addition, improper fusion strategies can lead to a reduction in the model's ability to detect small targets. While generative model-based denoising methods are effective, they inevitably increase model depth and computational complexity [18], [19], [20]. Therefore, we aim to explore the physical scattering mechanisms of SAR to develop auxiliary modules that eliminate various types of noise interference and accurately distinguish between background and targets, ultimately narrowing the visual–semantic gap.

B. Contribution

In this article, we propose a scattering-attention semi-supervised-guided reinforced region proposal (SSRP) network to tackle the abovementioned three problems and implement accurate ZSD tasks in SAR remote sensing images. To address and tackle the issue of data scarcity, we propose an engineering-level method, called $\mathcal{P}_2T$, specifically designed for generating textual data for ZSD, which can replace manual input and automatically produce key and comprehensive image descriptions with minimal expert knowledge. Furthermore, we enriched the SRSDD [21] dataset by adding detailed attribute labels and descriptions for each category. From the perspective of reinforced proposal for unseen classes and scattering noise interference, we propose the proposal optimization module (POM), which integrates knowledge distillation with adaptive knowledge distillation and self-reduction equations. This design enables the model to learn the symbiotic relationship between visual and semantic features, thereby strengthening cross-domain mapping connections and alleviating the weak response of proposal boxes to unseen classes. Furthermore, POM incorporates a scatter excitation learning module (SELM) that strategically activates neurons associated with local scattering centers during training. This allows the model to capture richer contextual and structural information without modifying the network architecture or inference mechanism, effectively mitigating the impact of scattering noise. Specifically, although the POM module is essentially a teacher–student framework based on traditional knowledge distillation, our innovation lies in tailoring it to the characteristics of SAR imagery. By incorporating additional SELM, the module addresses two key challenges in SAR-ZSD: the weak response bias toward unseen classes and the difficulty of robust visual–text feature interaction under severe speckle noise and structural ambiguity. To this end, we introduce enhanced components that jointly mitigate the weak response bias and strengthen visual–text feature interaction—capabilities that are absent in prior distillation frameworks. Therefore, the proposed POM can be regarded as a functional extension of the teacher–student paradigm, specifically adapted to the unique scattering characteristics and category-agnostic structural features of SAR images.

The highlights of our study are summarized as follows.

1) SSRP network is proposed for ZSD in SAR remote sensing imagery, which associates virtual reflection for characteristic domain from three perspectives, i.e., data augmentation, cross-domain feature transfer, and scattering characteristics. The SSRP network reduces the dependence of the SAR-ZSD task on large annotated datasets, enabling robust detection of unseen target classes.
2) To alleviate the scattering noise interference and weak response of ZSD to unseen classes, we propose the POM, which integrates knowledge distillation with adaptive knowledge distillation and self-reduction equations. In addition, it incorporates a scatter excitation learning mechanism, inspired by prior work [22], that leverages scattering point information between the scene and internal objects to emphasize object-correlated contexts, thereby strengthening visual–semantic correlation learning and suppressing irrelevant noise.
3) To address the scarcity of specialized SAR datasets for ZSD, we developed a dataset construction method that supports the automated conversion of images into representative descriptive text.

II. Related Work

As an extension and specific application of zero-shot learning (ZSL) in the field of object detection, ZSD incorporates the additional task of object localization, thereby presenting even greater challenges. Given the limited research on ZSD within the context of SAR imagery, we aim to provide a comprehensive review of existing ZSL studies and assess their limitations when applied to SAR-ZSD. We categorize these studies into three main approaches: 1) data manipulation and generative model-based methods; 2) label propagation-based methods; and 3) embedding-based methods.

A. Data Manipulation and Generative Model-Based Methods

Addressing the zero-shot problem directly or indirectly from the data level has always been the most intuitive approach. Data manipulation techniques, including detector-based methods, image cropping, and patch clustering, are employed to extract key local information from samples. Similar to local attention mechanisms, detector-based methods extract critical feature regions through multilayer convolutions. However, these approaches overly rely on detector performance and require accurate annotations for effective support [23], [24], [25]. A more effective strategy is to identify the optimal regions in seen class images and exclude irrelevant parts. Inspired by this concept, image cropping employs reinforcement learning and enhanced region embedding patterns, establishing well-defined reward objectives and serializing the task of searching for optimal regions to guide the model in identifying the most informative parts. Likewise, patch clustering exploits visual similarity, segmenting a set of images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

into clusters of local regions and learning their category distinctions and semantic correlations [26]. Nevertheless, these approaches often over-rely on optimization techniques and region preprocessing, where inaccurate cropping and clustering can substantially impair the semantic alignment accuracy for unseen classes.

Leveraging generative adversarial networks (GANs) [27] or variational autoencoders (VAEs) [28] to simulate unseen class samples and convert ZSD into traditional supervised learning represents another prominent direction for ZSD. Conventional generative model-based methods initially train a conditional generative model on seen classes using semantic descriptors and subsequently synthesize visual samples of unseen classes by combining semantic attributes. These synthesized samples are then combined with real samples from seen classes to train a supervised classifier. However, the accuracy of these approaches is often limited, as the distribution of synthetic samples does not follow the real data distribution, and the classifier has never encountered actual unseen class samples. Some studies have attempted to improve the generalization of mixed-class sample mapping by adding interclass decision boundaries in generative models or by integrating generative models with embedding models [29], [30], [31]. Focusing solely on global features can easily degrade the model's generalization and decision-making performance [32]. A better solution is to balance global and fine-grained knowledge within the generator. Researchers such as [33], [34], [35], [36], [37], and [38] have employed attention mechanisms or convolutional networks to direct the model's focus toward local regions, extracting and combining multiple key fine-grained features from these areas as realistic samples, thereby enhancing the generative capacity. Nevertheless, generative model-based methods are heavily dependent on high-quality semantic descriptions and prior knowledge. In the context of SAR imagery, defining attributes and descriptors can be challenging, often leading to incomplete or inaccurate category descriptions, which undermines the effectiveness of the generated samples in representing the target. Furthermore, the training process is computationally intensive and time-consuming, as it requires numerous iterations to optimize both the generator and the discriminator.

B. Label Propagation-Based Methods

In natural scenes, there may be some underlying relationships between the label information of known and unknown classes [39]. Graph structures are often used to represent real-world concepts or objects and their interactions. Graph convolutional networks (GCNs) capture and handle the complex relationships between seen and unseen classes by processing unstructured data, especially in scenarios where classes have potential semantic associations or neighborhood relationships. Assuming that $W^{(l)}$ represents the parameters of the lth layer in GCNs, the output of the $(l+1)$th layer can be expressed as follows:

$$F^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}F^{(l)}W^{(l)}\right) \tag{1}$$

where $\tilde{D}$ denotes its degree matrix and A is the adjacent matrix. σ denotes the activation function, and $F((l))$ is the output of the l-layer of GCNs. Inspired by this, Yan et al. [40] were the first to build a ZSD model based on graph networks, leveraging the semantic descriptions and structural knowledge presented in prior class graphs to enhance the generalization ability of the learned projection function. Aiming to explore the intrinsic relationships within attribute descriptions to obtain more discriminative attribute representations, APNet [41] employs graph modeling for attribute enhancement. As an improvement, the RGEN method [42] proposes representing the input image as a region graph, where each graph node corresponds to an attention region, calculated based on a single image and a single-layer convolutional feature map. Moreover, GKU [43] enhances the expression capabilities of key nodes in the graph structure through supervised learning with region annotations. Since label propagation does not require a large amount of sample data, these studies reduce the data demands for inferring unknown classes and effectively handle long-tail distributions. However, the weak focus of graph networks on certain features can result in unknown classes distant from known nodes receiving inaccurate or weakened labels. To address this issue, EOPA [44] combines graph networks with data manipulation techniques by constructing multigranularity hierarchical graphs to automatically identify region features associated with specific attributes. Since label propagation mainly relies on the exchange of local information between neighboring nodes, it is theoretically better suited for SAR image detection tasks, where information is sparse and global dependency is less critical compared to generative methods. However, the imaging characteristics of SAR depend on the geometric structure, scattering mechanisms, and physical properties of objects; the feature relationships become complex and unordered [45]. In the context of high-dimensional data and intricate nonlinear relationships in SAR image formation, simple label propagation may prove inadequate for capturing these complex associations.

C. Embedding-Based Methods

Embedding-based ZSL methods focus on learning embedding or mapping functions that associate the low-level visual features of seen classes with their corresponding semantic vectors. Notably, label propagation-based methods can also be broadly categorized as non-Euclidean space embedding-based methods, as they utilize geometric relationships between spaces to preserve the relationships between data samples. Thus, in this section, we emphasize related studies that do not involve graph networks.

Embedding-based methods can be categorized into three types based on their embedding approach: latent space embedding, visual embedding, and semantic embedding. Latent embedding involves mapping both visual and semantic features into a shared embedding space, where similarity matching or distance measurements are performed [46]. However, the low interpretability and high data requirements have hindered the development of latent space-based ZSL for SAR imagery. Visual and semantic embeddings employ forward projection from visual space to semantic space and backward projection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

from semantic space to visual space, respectively, to force the association between labels and test images. Visual embedding requires extensive labeled datasets, while semantic embedding relies more on prior knowledge or auxiliary information, which makes it more suitable for data-scarce SAR images. In semantic embedding methods, loss functions are often used to optimize the projection of additional information to address the unique properties of different modalities [47]. Improving task-specific loss functions was the initial motivation behind ZSL model advancements. Previous studies have explored the use of bilinear compatibility loss [48] and structured SVM loss [49] to learn the mapping between visual feature space and semantic space. The ideal loss function ensures intraclass compactness and interclass separability, a challenge that is particularly pronounced in low-resolution remote sensing images [50]. However, during training, seen classes tend to dominate the embedding space, leading to an inevitable issue of centrality. Li et al. [51] proposed an end-to-end approach that fully assimilates intramodal and intermodal supervision to alleviate the inconsistency in class structures between two mixed spaces. Similarly, the introduction of contrastive learning can help reduce the impact of centrality [52]. Compared to natural images, which require support for global visual features, ZSD for SAR imagery focuses more on specific target regions, which can also help mitigate data imbalance. In other words, embedding models enhanced with attention mechanisms enable the network to focus on learning the most important image regions, ultimately enhancing performance [53]. For instance, Huynh and Elhamifar [54] designed an attribute embedding technique that adjusts each derived attribute feature based on the corresponding semantic vectors, resulting in high-quality visual features. Additionally, semantic attention (SA) has been proposed to strengthen the element-wise correlation between vectors of seen and unseen classes [55]. These low-dimensional embedding mechanisms effectively filter out irrelevant features, thereby reducing the semantic–visual gap [56].

Compared to other approaches, embedding-based methods demonstrate superior performance in cross-task and cross-domain applications, making them particularly well-suited for model transfer from optical to SAR imagery. However, existing research on SAR imagery [57], [58], [59], [60] has predominantly focused on zero-shot classification, often adopting unsystematic approaches and providing a limited perspective, with much of the work centered around the MSTAR [61] dataset. Moreover, there is currently a lack of in-depth studies that address ZSD while accounting for the distinctive characteristics of SAR imagery.

III. Methodology

A. Overall Architecture

Fig. 2 illustrates the processing steps of the SSRP network. First, the POM consists of a teacher model, which provides continuous soft labels, and a student model, which captures detailed information. It generates weight information in a semi-supervised manner to enhance the network's ability to capture critical visual information in SAR images. Next, the feature extraction and encoding stage aims to map visual information to the attribute semantic matrix. The backbone network integrates the knowledge injected by the POM, enhancing its hierarchical feature extraction capabilities. Additionally, the scatter excitation learning module eliminates SAR noise interference, allowing the model to focus on the foreground regions. Similarly, the POM guides the RPN, improving its ability to locate unseen target regions and generate proposal boxes. Subsequently, the region of interest (RoI) pooling layer and multiple fully connected layers jointly process the initial feature mappings and proposal boxes to generate visual feature vectors for feature alignment. Furthermore, we propose a dataset construction method, $\mathcal{P}_2T$, specifically designed for the SAR-ZSD task, which supports the extraction of textual attribute semantic information from unseen images. Finally, in the visual-attribute semantic mapping, visual features are mapped to the attribute semantic matrix of unseen classes via similarity comparison, enabling cross-domain detection from seen to unseen classes. The loss function for the network is composed of a proposal optimization loss focused on feature extraction, as well as the box regression loss and max-margin loss, which focus on training feature alignment capabilities.

B. Data Format Definition

We first provided a detailed definition of the data format. Under the SAR-ZSD task, the datasets comprise both unseen classes and seen classes. We represent the parameters as follows:

$$\begin{aligned} S &= \left\{c_i^s \,|i = 1, 2, 3, \ldots, N_s\right\} \\ U &= \left\{c_i^u \,|i = 1, 2, 3, \ldots, N_u\right\} \end{aligned} \tag{2}$$

where S represents the set of seen classes and the unseen classes encapsulated by the set U and c_i^s and c_i^u denote a seen class and an unseen class, respectively. The unseen classes are only used to generate attribute matrices for semantic mapping during the visual-attribute semantic mapping stage and are not directly input into the network for training images. The seen classes will first be randomly assigned labeled and unlabeled tags and input into the POM to train the semi-supervised weights. Once the task is completed, the labels of the seen classes will be removed, and both the weights and the seen classes will be fed into the backbone network as input for the feature extraction and encoding task. The training dataset D^{tr} can be represented as follows:

$$D^{\text{tr}} = \left\{\left(x_i^{\text{tr}}, y_i^{\text{tr}}, p_i^{\text{tr}}\right) \subseteq \mathcal{X} \times S \times P\right\}. \tag{3}$$

Here, x_i^{tr} is the input representation in feature space obtained from pretrained deep learning models and these images contain at least one instance of seen class. y_i^{tr} constitutes the labels for instances enveloped by (x_i^{tr}, and p_i^{tr}) delineates the set of bounding boxes for the y_i^{tr} Typically, each bounding box is defined by a rectangular area determined by coordinates ($x_{\min}, y_{\min}, x_{\max}$, and $y_{\max}$). The D-dimensional feature space $\mathcal{X}$, typically a subset of $\mathbb{R}^D$, consists of data mapped by networks such as ResNet and VGG. The testing dataset X^{te} contains testing images in feature space

$$X^{\text{te}} = \left\{\left(x_i^{\text{te}} \in \mathcal{X}\right\}_{i=1}^{N\text{te}}. \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

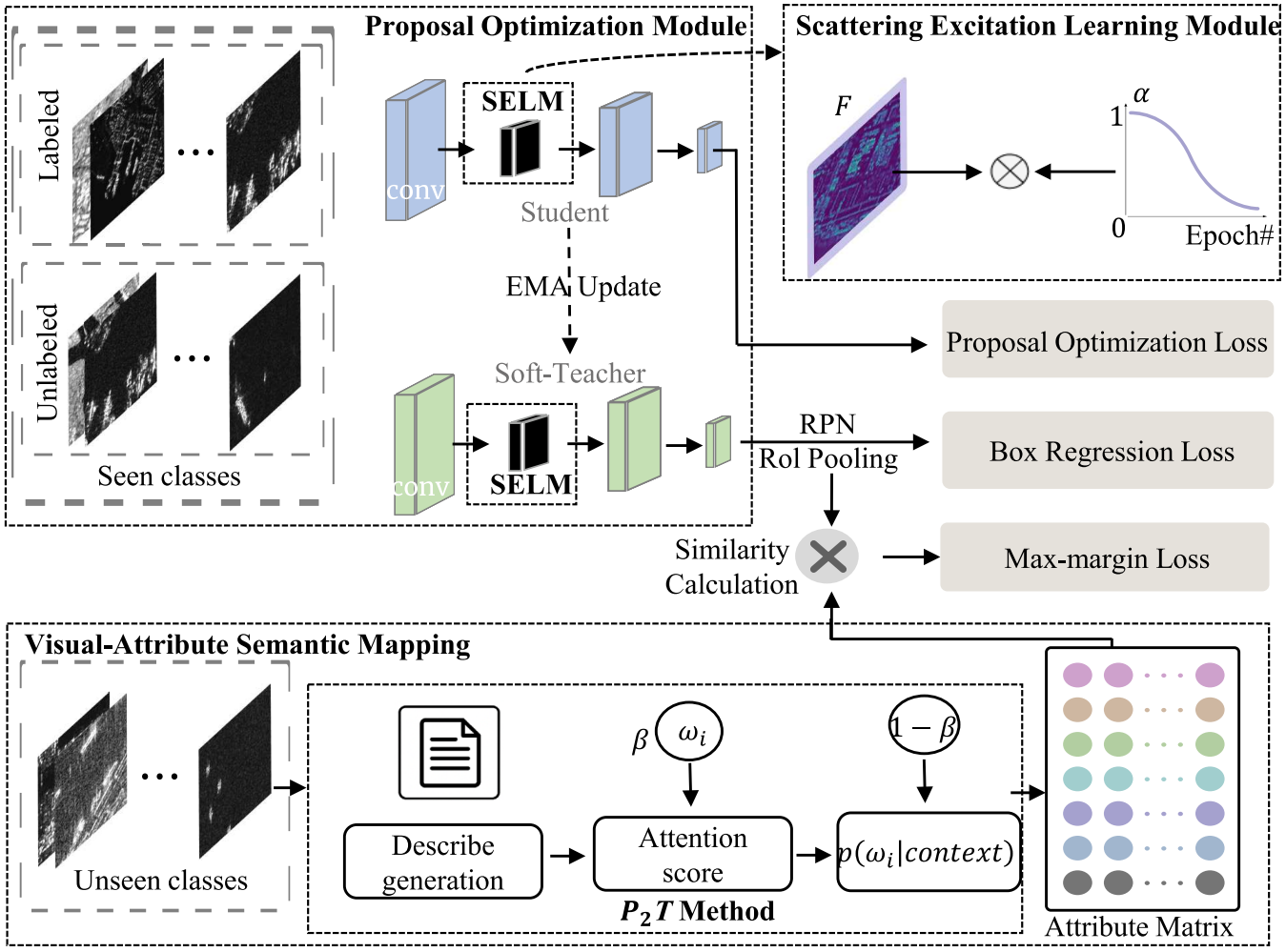


Fig. 2. Overview of the proposed framework.

While no instances of unseen classes are present within D^{tr}, the set X^{te} might include objects from seen classes.

C. Proposal Optimization Module

We identify that the key reason for low accuracy in recognizing unseen classes in the SAR-ZSD task is the lack of visual information in SAR images, which increases the difficulty of cross-domain knowledge transfer and alignment in models. This issue is particularly pronounced when labeled data are limited, as traditional two-stage ZSD models suffer from feature misalignment problems, leading to nonresponse of proposal boxes to unseen classes. To address these challenges, we propose the POM, which aims to provide additional visual features to enhance the model's understanding of interclass diversity and intraclass variability, thereby expanding the feature distribution coverage. Specifically, the POM optimizes the backbone and region proposal network (RPN) by employing a soft-teacher model to improve the representation capability of the input data, as shown in Fig. 3. Following the mainstream consistency regularization semi-supervised approach, the model is trained on both labeled and unlabeled images. The

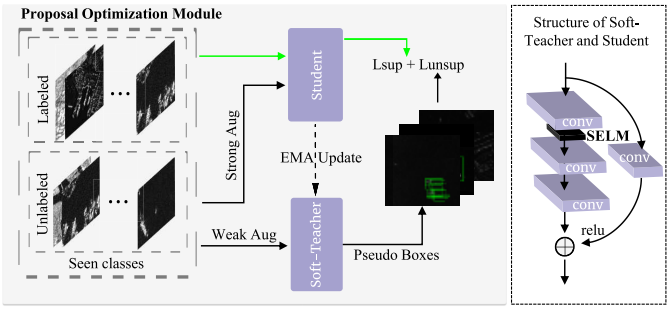


Fig. 3. Structure of the POM.

objective of the POM is to train a model for visual-attribute semantic mapping

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \alpha \mathcal{L}_{\text{unsup}} \tag{5}$$

where $\mathcal{L}_s$ and $\mathcal{L}_{\text{unsup}}$ represent the supervised loss for labeled images and the unsupervised loss for unlabeled images, respectively, and α controls the contribution of the unsupervised loss. Both losses are normalized according to the respective number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

of images in the batch

$$\mathcal{L}_{\text{sup}} = \frac{1}{N_l}\sum_{i=1}^{N_l}\left(\mathcal{L}_{\text{cls}}\left(I_l^i\right) + \mathcal{L}_{\text{reg}}\left(I_l^i\right)\right) \tag{6}$$

$$\mathcal{L}_{\text{unsup}} = \frac{1}{N_u}\sum_{i=1}^{N_u}\left(\mathcal{L}_{\text{cls}}\left(I_u^i\right) + \mathcal{L}_{\text{reg}}\left(I_u^i\right)\right) \tag{7}$$

where I_l^i denotes the ith labeled image; I_u^i represents the ith unlabeled image; $\mathcal{L}_{\text{cls}}$ is the classification loss; $\mathcal{L}_{\text{reg}}$ refers to the box regression loss; and N_l and N_u signify the number of labeled and unlabeled images, respectively. By incorporating unlabeled samples, the POM effectively broadens the feature distribution and captures intraclass variability and interclass diversity, which are crucial for improving the recall in zero-shot scenarios. In particular, the semi-supervised consistency learning encourages the RPN to generate proposals that are more likely to cover potential unseen class objects, thereby reducing missed detections (false negatives) and enhancing recall. Unlike standard teacher–student setups designed for general knowledge transfer, our POM is the first to integrate semi-supervised learning with cross-modal alignment, specifically for SAR-ZSD. We fundamentally redefine the teacher's role to jointly mitigate weak response bias toward unseen classes and enhance visual–text feature interaction—a capability absent in prior distillation frameworks. Meanwhile, our framework unifies distillation with adaptive feature enhancing and self-reducing equation, constituting novel architectural contributions tailored for open-world SAR-ZSD scenarios.

D. Scattering Excitation Learning Module

To effectively leverage the scattering characteristics inherent in SAR imagery, the SELM is strategically positioned following the first convolutional layer of the backbone network. This deliberate placement is motivated by two key considerations: 1) early stage feature maps retain richer low-level scattering patterns that are crucial for SAR target interpretation and 2) injecting physical scattering priors at this stage allows subsequent layers to progressively build more discriminative representations. The module's location ensures optimal preservation of critical scattering signatures while mitigating the interference of speckle noise in deeper network layers. As shown in Fig. 4, this architectural design enables the model to maintain scattering feature purity throughout the hierarchical feature extraction process, particularly beneficial for ZSD scenarios where generalization to unseen classes relies heavily on robust fundamental feature representation. The SELM takes an input feature map $\vec{F} = [f_{(c,i,j)}] \in \mathbb{R}^{C\times H\times W}$ and outputs the feature map $\hat{\vec{F}} \in \mathbb{R}^{C\times W\times H}$. The processing can be described

$$\hat{\vec{F}}^{(c)} = \vec{F}^{(c)} + \alpha(t)\,\vec{E} \tag{8}$$

where $\hat{\vec{F}}^{(c)}$ and $\vec{F}^{(c)}$ are the cth channel of the feature map $\hat{\vec{F}}$ and $\vec{F}$, respectively, and $\vec{E} = [e_{(i,j)}]$ is a scattering-attention feature map generated by first averaging the feature map $\vec{F}$ along its channel dimension and subsequently performing

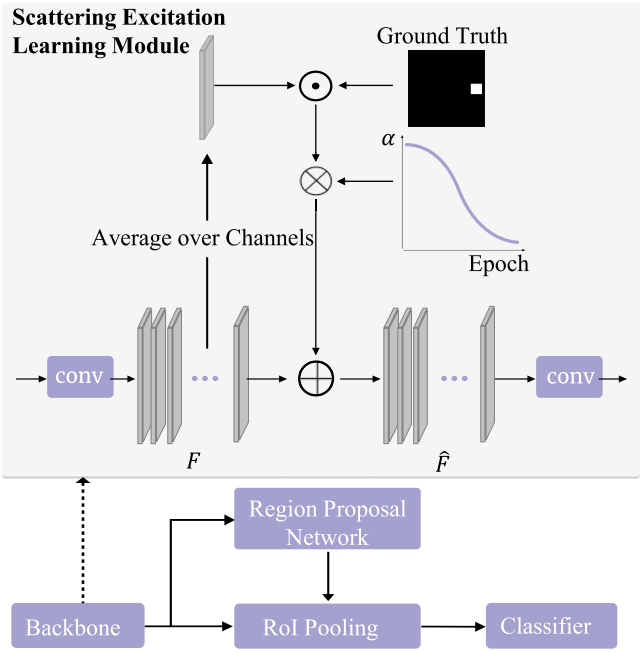


Fig. 4. Structure of the scatter excitation learning module.

an element-wise multiplication with the ground-truth map $\vec{G} = [g_{(i,j)}]$, i.e.,

$$e_{(i,j)} = g_{(i,j)} \cdot \frac{1}{C}\sum_{c=1}^{C} f_{(c,i,j)} \tag{9}$$

where $\vec{G} = [g_{(i,j)}]$ is produced by bounding box of ground truth. If $g_{(i,j)}$ is within the bounding box, it is assigned a value of 1; otherwise, it is set to 0. $\alpha(t)$ is a modulation factor dependent on epoch t, which is initialized with a nonzero value during the first epochs and gradually decreases to zero during training. It is calculated as follows [22], [62]:

$$\alpha(t) = \frac{1}{2} \times \frac{1 + \text{Cos}(\pi \cdot t)}{\text{MaxIteration}}. \tag{10}$$

Note that the SELM module is only applied at the training stage and is not employed in the inference stage, i.e., $\alpha(t) = 0$ at the testing stage. Through this approach, our model progressively learns the target scattering characteristics while autonomously operating on the ground-truth scattering centers, thereby enhancing feature extraction robustness.

E. SAR-ZSD Dataset Construction Method

One of the primary obstacles to applying SAR-ZSD is the lack of comprehensive and reliable SAR datasets suitable for ZSD. In terms of data annotation, unlike optical images, SAR images suffer from a lack of information and limitations in color, fine texture, and scene comprehension. This makes it challenging to expand the interclass gap in textual descriptions. We developed a process and method for constructing SAR zero-shot datasets, called $\mathcal{P}_2T$, to facilitate the development of SAR-ZSD. $\mathcal{P}_2T$ is designed to integrate the entire SAR-ZSD dataset based on attribute information and convert the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT BACKBONES ON THE MSTAR DATASET

Method	Occlusion Level / Direction	0% -d1	0% -d3	10% -d1	10% -d3	20% -d1	20% -d3	30% -d1	30% -d3	40% -d1	40% -d3	50% -d1	50% -d3
A-ConvNet		99.46	99.46	98.89	98.92	96.49	96.74	88.74	89.81	70.92	73.40	49.07	47.96
A-ConvNet+SELM		**99.67**	**99.67**	**99.17**	**99.34**	**97.69**	**98.27**	**93.11**	**93.03**	**80.08**	**81.24**	**60.79**	**60.58**
VGG		97.07	97.07	**96.29**	96.49	92.91	93.69	82.14	81.61	60.95	59.71	38.43	38.52
VGG+SELM		**97.57**	**97.57**	95.67	**96.75**	**95.05**	**94.88**	**88.57**	**87.18**	**75.67**	**73.81**	**58.85**	**57.11**
ResNet		**95.79**	**95.79**	95.18	95.09	92.95	93.07	83.30	82.76	64.66	65.94	46.68	49.40
ResNet+SELM		95.38	95.38	**95.26**	**95.29**	**93.90**	**94.19**	**88.16**	**88.04**	**76.82**	**77.03**	**61.24**	**61.81**

TABLE II

EXPERIMENTAL RESULTS OF SELM PLACED AT THE DIFFERENT POSITIONS OF THE NETWORK ON THE MSTAR DATASET

Method \ Occlusion level	0	10%	20%	30%	40%	50%
A-ConvNet (SELM@Stage 1)	**99.67**	**99.29**	**98.04**	**93.76**	**82.54**	**60.66**
A-ConvNet (SELM@Stage 2)	99.46	98.92	96.88	89.65	72.37	49.24
VGG (SELM@Stage 1)	**97.57**	97.24	**95.63**	**88.16**	**71.17**	**52.57**
VGG (SELM@Stage 2)	96.95	**97.32**	94.89	84.70	65.32	45.48
ResNet (SELM@Stage 1)	95.38	95.26	**93.90**	**88.16**	**76.82**	**61.24**
ResNet (SELM@Stage 2)	**95.95**	**95.38**	92.78	82.80	64.70	44.61

TABLE III

MAP OF SELM ON SRSDD

Methods	Seen					Unseen		
	Container	Dredger	Fishing	LawEnforce	Overall	Ore-oil	Cell-Container	Overall
ResNet	51.07	**53.65**	52.21	63.90	55.21	3.70	9.59	6.64
ResNet+SELM	**51.47**	50.51	**59.19**	**78.79**	**59.99**	**9.25**	**10.82**	**10.04**

target objects in each image into multiple representative descriptors. We propose using standardized annotations rather than crowdsourcing to avoid limitations in annotation quality diversity and to increase scalability. Characteristics such as target scattering and motion are particularly prominent in SAR images. Therefore, $\mathcal{P}_2T$ primarily focuses on structural, motion, and attribute information of the target objects, aiming to distinguish as many target categories as possible with the most concise vocabulary. We regard $\mathcal{P}_2T$ as a tool for generating textual descriptions and performing attribute-based filtering tasks. $\mathcal{P}_2T$: The API of OpenAI GPT-4 technology is used to generate an initial detailed description of the visual image corresponding to the jth category in the SAR image. This description is as comprehensive as possible, thus containing a large number of redundant sentences. The next step is to retain the core semantic features for each category. We calculate the average attention weight of each word using the BERT [63]

$$\text{Attention_score}\,(w_i) = \frac{1}{n}\sum_{j=1}^{n} a_{ij} \tag{11}$$

where a_{ij} represents the attention weight of word w_i. Next, all words in the sentence are ranked based on their importance scores, and we attempt to remove those that have minimal impact on the overall sentence semantics. The contribution of each word can be evaluated using the following formula:

$$\text{Score}\,(w_i) = \beta \cdot \text{Attention_score}\,(w_i) + (1-\beta)\cdot p\,(w_i \mid \text{context}) \tag{12}$$

where β is a weight coefficient and $p(w_i||\text{content})$ represents the contextual reconstruction probability, specifically the model's prediction probability of word w_i given the context. By combining attention weights and contextual reconstruction probabilities, we comprehensively evaluate the redundancy of a word. If Score(w_i) is below a certain threshold, the word is considered redundant and can be removed. We directed ChatGPT to focus on generating descriptions that emphasize the intrinsic characteristics of the target object, including its structural features, motion patterns, and attribute-specific information (e.g., appearance characteristics, motion information, and functional properties). Descriptions were tailored to capture key attributes of the target category without cross-category comparisons, avoiding unnecessary bias or contamination from other classes. In the SAR-ZSD data construction process, expert opinions are incorporated to manually review the automatically generated descriptions, ensuring minimal loss of attribute matrix and descriptor information while meeting the dataset's requirements. Although $\mathcal{P}_2T$ does not introduce a novel algorithm, it represents a task-specific, performance-critical methodological design tailored to the SAR ZSD setting, and plays a pivotal role in enabling effective visual–semantic alignment under limited supervision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RECALL@100 FOR POM ON SRSDD

Methods	Seen					Unseen		
	Container	Dredger	Fishing	LawEnforce	Overall	Ore-oil	Cell-Container	Overall
ResNet	**78.78**	72.09	52.44	83.33	71.66	35.95	63.04	49.50
ResNet+POM	75.10	73.52	75.18	94.44	79.56	**47.40**	87.50	67.45
ResNet+SELM+POM	78.05	**75.10**	**78.28**	**100.00**	**82.86**	47.19	**89.13**	**68.16**

TABLE V
MAP OF SEMI-SUPERVISED ON SRSDD

Methods	Seen					Unseen		
	Container	Dredger	Fishing	LawEnforce	Overall	Ore-oil	Cell-Container	Overall
ResNet	51.07	53.65	52.21	63.90	55.21	3.70	9.59	6.64
ResNet+POM	**57.56**	51.30	53.58	91.58	63.50	10.30	21.42	15.84
ResNet+SELM+POM	56.3	**54.2**	**56.7**	**97.6**	**66.2**	**15.39**	**27.78**	**21.59**

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS ON THE SRSDD DATASET

Methods	Seen/Unseen	Recall@100			mAP
		0.4	0.5	0.6	0.5
BLC [65]	4/2	7.12	6.56	4.38	2.0
ZSI [66]	4/2	50.0	48.9	27.2	5.5
WEA-DINO [67]	4/2	47.8	47.8	41.3	6.0
SUZOD [68]	4/2	**76.6**	65.0	53.5	11.8
ZSD-SCR [69]	4/2	68.98	59.61	48.60	14.7
RRFS [70]	4/2	74.8	64.0	51.7	17.1
Ours	4/2	70.33	**68.16**	**64.30**	**21.59**

Based on the aforementioned method, we employ the SRSDD [21] dataset for SAR-ZSD. The SRSDD consists of 666 images cropped from 30 original panoramic images. Among these, 420 images depict land coverage, with 2275 detected ships. Additionally, there are 246 images featuring only ocean backgrounds, containing 609 ships. Coastal scenes make up 63.1% of the dataset, while open ocean scenes constitute 36.9%. The dataset encompasses six ship types: ore-oil ships, bulk cargo ships, fishing boats, law enforcement ships, dredger ships, and container ships.

The descriptors shown in Table VII primarily describe different ship categories from the perspectives of functional attributes, image attributes, and appearance characteristics, capturing the essence of each category from top to bottom. Before generating the final descriptors, we filtered out most attributes with excessive potential overlap or heterogeneity, refining the attribute matrix to be as concise as possible, as shown in Table VIII. Specifically, descriptors such as "Large Size" and "Smaller Vessel" reflect the appearance characteristics of ship objects. "Multipurpose Use" and "Operational for Transport" mainly distinguish the functional attributes of different ships, while "Capable of High Speed" and "High Radar Cross Section" highlight the scattering characteristics and motion information unique to SAR images compared to optical images. Some descriptors maximized the intraclass distance of ships; for example, "Underwater Operation Capable" effectively distinguishes dredger ships from the other six categories. Similarly, only dredger ships possess the "Frequent Shore Operations" attribute. $\mathcal{P}_2T$ ensures a certain degree of correlation between interclass attributes to provide sufficient support for transferring features from seen to unseen categories. Attributes like "Specialized Equipment" and "High Radar Cross Section" are included in the descriptions of four ship categories. It is important to note that certain descriptors, such as large or small ship size, or fast or slow speed, are not universally applicable. This is because, in natural scenarios, it is not guaranteed that all samples will lack heterogeneity.

In addition, one should note that the database construction in our work was not intended as a methodological innovation but rather as a necessary step to evaluate the proposed framework. To this end, we provided a practical strategy for building the SAR-ZSD dataset, which relies on pretrained language models such as GPT and BERT to generate textual descriptions of ship categories, followed by manual screening to remove redundant words and obtain semantic vector representations. Although this process indeed reflects more of an engineering-level operation rather than a methodological breakthrough, it should be emphasized that this is not the primary contribution of the article, but only a feasible reference for constructing SAR-ZSD datasets. Moving forward, we plan to develop more automated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
DESCRIPTIONS OF SRSDD

bulk cargo ship	Bulk cargo ships are large vessels designed to transport a variety of bulk materials, such as grains, minerals, and coals. These ships have a high radar cross-section and a streamlined, single-hull design with large hatches on fewer decks, optimizing them for loading and unloading bulk materials. The deck load capacity of these ships is significant, allowing them to transport heavy and large quantities of cargo. While these vessels do not feature complex superstructures or container stacks, their operational design emphasizes the efficient movement of bulk goods, often navigating to and from industrial ports. Bulk cargo ships typically do not have the high-speed capabilities or specialized equipment seen on more specialized vessels but are essential for the global transportation of raw materials, playing a critical role in numerous industrial supply chains.
dredger ship	Dredgers are medium-sized, specialized vessels designed for underwater operations such as excavation and removal of sediment from waterbeds. They are equipped with heavy machinery tailored for dredging activities, which often requires dynamic positioning systems to maintain precise locations during operations. Dredgers have a unique hull design that accommodates dredging equipment and often features a higher width-to-length ratio to provide stability during operations. The superstructure on dredgers is variable, primarily functional, and designed to support dredging operations. They are not built for high speed but are capable of frequent shore operations, which is essential for their role in harbor and waterway maintenance. Dredgers often operate close to shore and are critical in projects that require alteration of the water environment, such as channel deepening and harbor clearing. These vessels contribute significantly to maritime infrastructure maintenance, ensuring navigational and operational safety for other vessel types.
fishing boat	Fishing boats are small, compact vessels designed specifically for stability and efficiency in various marine environments. These boats are equipped with specialized fishing equipment, including nets and lines, and are often characterized by colorful appearances and lower radar cross-sections in SAR imagery. Despite their small size, fishing boats are built to withstand rough waters and are capable of high-speed operations to chase schools of fish or return quickly to shore. They frequently operate close to shore and are integral to the coastal economy. Fishing boats have simple, functional decks and lack the complex superstructures seen in larger commercial vessels. Their design maximizes efficiency and effectiveness in capturing seafood, supporting both commercial and subsistence fishing activities.
law enforcement ship	Law enforcement ships are medium-sized vessels equipped with advanced technologies, including radars and specialized law enforcement equipment. They have a high radar cross-section and a streamlined design for rapid response capabilities. Enforcement ships are characterized by white or grey colors with official markings to denote their authority. These vessels are designed with ice-class reinforcements to operate in diverse marine environments, including icy waters. They feature enhanced safety features and a prominent navigation bridge for optimal visibility and control during operations. Enforcement ships are capable of high-speed maneuvers and are equipped with pollution control equipment to address environmental concerns during maritime law enforcement operations. These ships play a crucial role in maintaining maritime security, enforcing regulations, and conducting search and rescue operations, making them essential assets for governmental maritime authorities..
container ship	Container ships are among the largest maritime vessels, designed specifically for transporting containers stacked in clear, visible layers. These ships feature a very high radar cross-section and complex superstructures, accommodating multiple accessible decks and a prominent navigation bridge. Container ships are characterized by their double-hull design, which helps in handling the substantial load of containers. Equipped with cranes and other heavy machinery, these vessels are operational powerhouses, capable of high deck load capacity. Their dynamic positioning systems and modular designs make them versatile and essential for global commerce. The container ship's design focuses on operational efficiency, safety, and the capacity to handle a large crew, reflecting the extensive human resources required for their operations.
ore-oil ship	Ore-oil ships are large, multifunctional vessels designed for transporting oil and ore, featuring high tonal intensity and a high radar cross-section in SAR imagery. These ships typically have a streamlined, single-hull design, enhancing their operational efficiency for long maritime voyages. The deck configuration is minimal, focusing on maximizing cargo space rather than complex superstructures, which are often smaller and less prominent compared to other large vessels. Despite their large size, ore-oil ships are not equipped for high-speed operations; their primary focus is on safe, efficient transport of bulk liquid and solid cargoes. These vessels are also equipped with pollution control equipment, underscoring a commitment to environmental safety standards. The ore-oil ship's design includes features to minimize environmental impact while ensuring the safe transportation of potentially hazardous materials. This type of vessel lacks container stacks and heavy machinery, indicating its specialized cargo handling requirements that differ from those of container or bulk cargo ships.

and learning-based semantic optimization mechanisms, which will not only reduce manual intervention but also enhance the scalability and methodological rigor of our framework, thereby enabling further breakthroughs as SAR-ZSD research progresses.

F. Loss Function

In our implementation, we used an equal number of positive and negative proposals. Now, as object proposals pass through ROI-Pooling and subsequent dense layers, the feature representation fi of each ROI is computed, which is then passed to two branches: the classification branch and the regression branch. The overall loss is the sum of the corresponding losses from these two branches, namely, the classification loss and the bounding box regression loss

$$L\left(\mathbf{o}_i, b_i, y_i, b_i^*, \theta\right) = \frac{1}{\mathrm{T}} \sum_i \left(L_{\mathrm{c\,1\,s}}\left(\mathbf{o}_i, y_i\right) + L_{\mathrm{reg}}\left(b_i, b_i^*\right)\right) \quad (13)$$

where θ denotes the parameters of the network; $\mathbf{o}_i$ is the classification branch output; $T = N \times R$ represents the total number of ROIs in the training set with N images; b_i and b_i^* are parameterized coordinates of predicted and ground-truth bounding boxes, respectively; and y_i represents the true class label of the ith object proposal. For the classification branch, it is essential to consider all categories within C', encompassing seen classes, unseen classes, and background. The loss function from [64] is adopted

$$L_{\mathrm{cls}}\left(o_i, y_i\right) = \frac{1}{|C' \backslash y_i|} \sum_{c \in C' \backslash y_i} \log\left(1 + \exp\left(o_c - o_{y_i}\right)\right). \quad (14)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
ATTRIBUTE MATRIX OF SRSDD

Categories \ Attributes	At_1	At_2	At_3	At_4	At_5	At_6	At_7	At_8	At_9	At_10	At_11	At_12	At_13	At_14
bulk cargo ship	0	0	0	1	1	0	1	0	1	0	1	0	1	0
dredger ship	1	0	1	0	0	1	0	1	0	1	0	1	0	1
fishing boat	1	0	1	0	0	1	0	0	0	1	0	1	0	1
law enforcement ship	1	1	1	0	0	0	0	0	0	1	0	0	1	0
container ship	1	0	0	1	1	0	1	0	1	0	1	0	1	0
ore-oil ship	0	1	0	1	1	0	1	0	1	0	1	0	1	0
background	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The detailed attributes are:
^{At_1} Specialized Equipment
^{At_2} Multi-Purpose Use
^{At_3} Other Operational
^{At_4} Large Size
^{At_5} Distinct Wake
^{At_6} Smaller Vessel
^{At_7} Operational for Transport
^{At_8} Underwater Operation Capable
^{At_9} High Tonal Intensity
^{At_10} Capable of High Speed
^{At_11} Deck Load Capacity
^{At_12} Frequent Shore Operations
^{At_13} High Radar Cross Section
^{At_14} Color Variation

The part of regression loss is the same as the faster-RCNN regression loss.

IV. EXPERIMENTS

A. Experimental Setting

1) Dataset: We employ two distinct datasets: MSTAR [61] and SRSDD [21]. An ablation study of SELM and POM experiments is conducted on the MSTAR dataset to demonstrate the robustness of our method in SAR target recognition. Additionally, we perform experiments on the SRSDD dataset to validate our approach for SAR-ZSD.

1) *MSTAR:* Due to the relatively simple structure of the dataset, in order to enhance the experimental challenge and data complexity, we processed the dataset using the occlusion method proposed in [22]. This method simulates potential occlusions that might occur in actual battlefield environments, thereby increasing the difficulty of target recognition and more effectively evaluating the robustness of our method.
2) *SRSDD:* Among six object categories, we select two categories (ore-oil ships and container ships) as unseen and the rest of the four categories (bulk cargo ships, fishing boats, law enforcement ships, and dredger ships) are considered as seen, following similar rules to [64]. The modified training set comprises 471 images with a total of 2244 annotated bounding boxes. For testing purposes, we employ the conventional zero-shot recognition setting, which exclusively only focuses on unseen classes. Therefore, we select the test data from the dataset that only includes the object categories of ore-oil ships and container ships. The resulting test set has 111 images and 135 bounding boxes.

2) Evaluation Metric: We calculated the average precision (AP) for each seen/unseen class and the mean AP (mAP) to assess the cumulative performance across all seen/unseen classes. We chose this particular evaluation metric because of its extensive adoption in the realm of supervised object detection tasks, where it serves as a standard measure of an algorithm's precision at various threshold levels, offering a comprehensive view of its detection capabilities. Furthermore, we employed the recall metric to assess the recall rate for unseen classes, thereby validating the effectiveness of our approach. The recall evaluation was based on Recall@K, which measured the recall rate when only the top K detections from an image were selected. Following the configuration in [8], we set K at 100.

3) Implementation Details: We implemented the proposed model based on the open-source package MMDetection. In Step 1, we trained the model with two GPUs for 180 000 iterations with an SGD optimizer whose momentum is 0.9 and weight decay is 0.0001. The initial learning rate for the optimizer is set to 0.01 and decreased by 0.1 after 120 000 and 160 000 iterations. In Step 2, we set the parameters broadly consistent with those in Step 1, training for a total of 36 000 iterations, with the learning rate decreased by a factor of 0.1 after 20 000 and 30 000 iterations.

B. Ablation Studies

In this section, we describe how we conducted comprehensive experiments to analyze the proposed modules in the SSRP network. In addition to the conventional ablation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

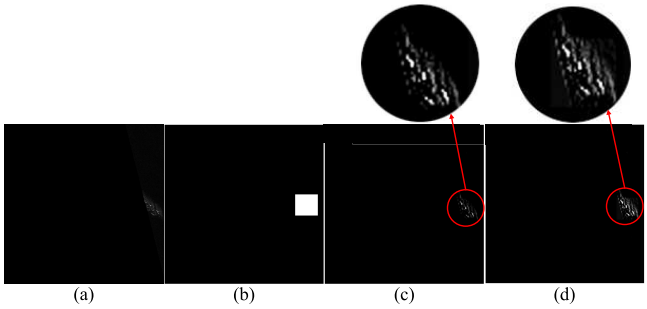


Fig. 5. Visualization of the feature response maps. (a) Original image. (b) Ground-truth mask of SELM. (c) Feature map without SELM. (d) Feature map with SELM.

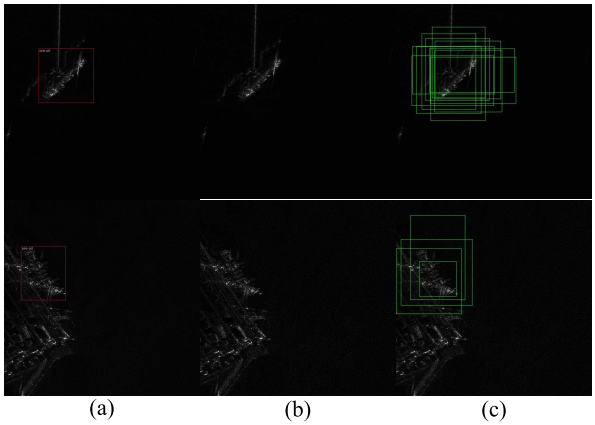


Fig. 6. Visualization of the proposal bounding box. (a) Ground truth for unseen classes. (b) Proposal for unseen classes of supervised method. (c) Proposal for unseen classes of semi-supervised method.

self-comparison experiments, we also modified the backbone network and the position to investigate the transfer ability of the SELM and the POM module across different datasets. To ensure a fair validation of the method's effectiveness, all other parameters in the experiment were kept unchanged except for adding our proposed method.

1) Scattering Excitation Learning Module: To investigate the robustness of the SEML module in feature extraction, we conducted performance evaluations using synthetically occluded SAR images. Adopting the occlusion strategy from [45], we sequentially removed the top 200 strongest scattering centers by zeroing their pixel values when they aligned with any of four predefined directions—azimuth (d_1), range (d_2), reverse azimuth (d_3), or reverse range (d_4)—and belonged to the dominant scattering set. This iterative process continued until the desired occlusion ratio was achieved. Given that azimuthal occlusion (d_1/d_3) comprehensively reflects target anisotropy in SAR imagery without experimental redundancy, we focused validation on these two directions. The results are summarized in Table I. Additionally, we have included various image occlusion conditions for each direction as additional comparative items to enrich our experiments. For instance, 0% represents no occlusion, while 50% indicates that half of the target is occluded. When the occlusion rates are 0% and 10%, the improvement brought by SELM to the network is not significant, amounting to only about 0.5%. As the occlusion rate increases, the image features become more blurred, and their representations become more similar to those of unseen classes, making the role of SELM increasingly pronounced. When the occlusion rate reaches 50%, SELM can enhance the classification accuracy of A-ConvNet, VGG, and ResNet by 12.62%, 19.55%, and 12.41%, respectively.

In addition, we also investigated the optimal positioning of the SELM within various network architectures, as illustrated in Table II. Specifically, Stage 1 represents placing the SELM after the first convolutional block, while Stage 2 positions it after the second convolutional block. It can be observed that regardless of its position, the SELM helped improve the recognition accuracy of both backbone networks. Except for the case with 10% occlusion, placing the SELM in Stage 1 resulted in higher accuracy than in Stage 2 for all other occlusion levels.

In the SSRP network, the SELM was positioned in Stage 1 to enhance the details of low-dimensional scattering features and mitigate the interference from speckle noise. Fig. 5 visualizes the feature maps before and after the addition of SELM. It is evident that the feature map in Fig. 5(d) contains more details and is clearer than the one in Fig. 5(c). Notably, the SELM is applied only during the training process and does not introduce any additional burden during other stages.

We further validate our approach on the SRSDD dataset, where the detection accuracy for each category within both seen and unseen classes is detailed, as shown in Table III. In the context of the SAR-ZSD task, SELM demonstrates a more powerful supporting role. The overall accuracy of seen classes has increased by an average of 4.78%, and the accuracy of the Lawenforce class has even improved by 14.89%. Similarly, the accuracy of unseen classes has also increased comprehensively by approximately 3.4%.

2) Proposal Optimization Module: To investigate the effectiveness of POM in SAR-ZSD, we conducted ablation comparison experiments on the SRSDD dataset. Table IV displays the outcomes of POM in terms of recall@100. The POM module significantly boosts the model's recall for seen classes and also markedly enhances its ability to recognize unseen classes. Notably, compared to the ore-oil class, the cell-container class approach demonstrates a significant improvement in performance. With the addition of SELM and POM, the performance further increases to 89.13%. In particular, POM introduces a semi-supervised approach with soft thresholds, which markedly enhances the proposal capability of the backbone network for unseen classes, achieving an average improvement of 18.66%. In comparison to SELM, POM contributes more significantly to intraclass domain adaptation for both features and bounding boxes in ZSD tasks.

To intuitively demonstrate the role of POM, we have visualized the process of semi-supervised proposal box generation. Fig. 6 presents a comparison of results using supervised and unsupervised methods. It can be observed that by incorporating

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

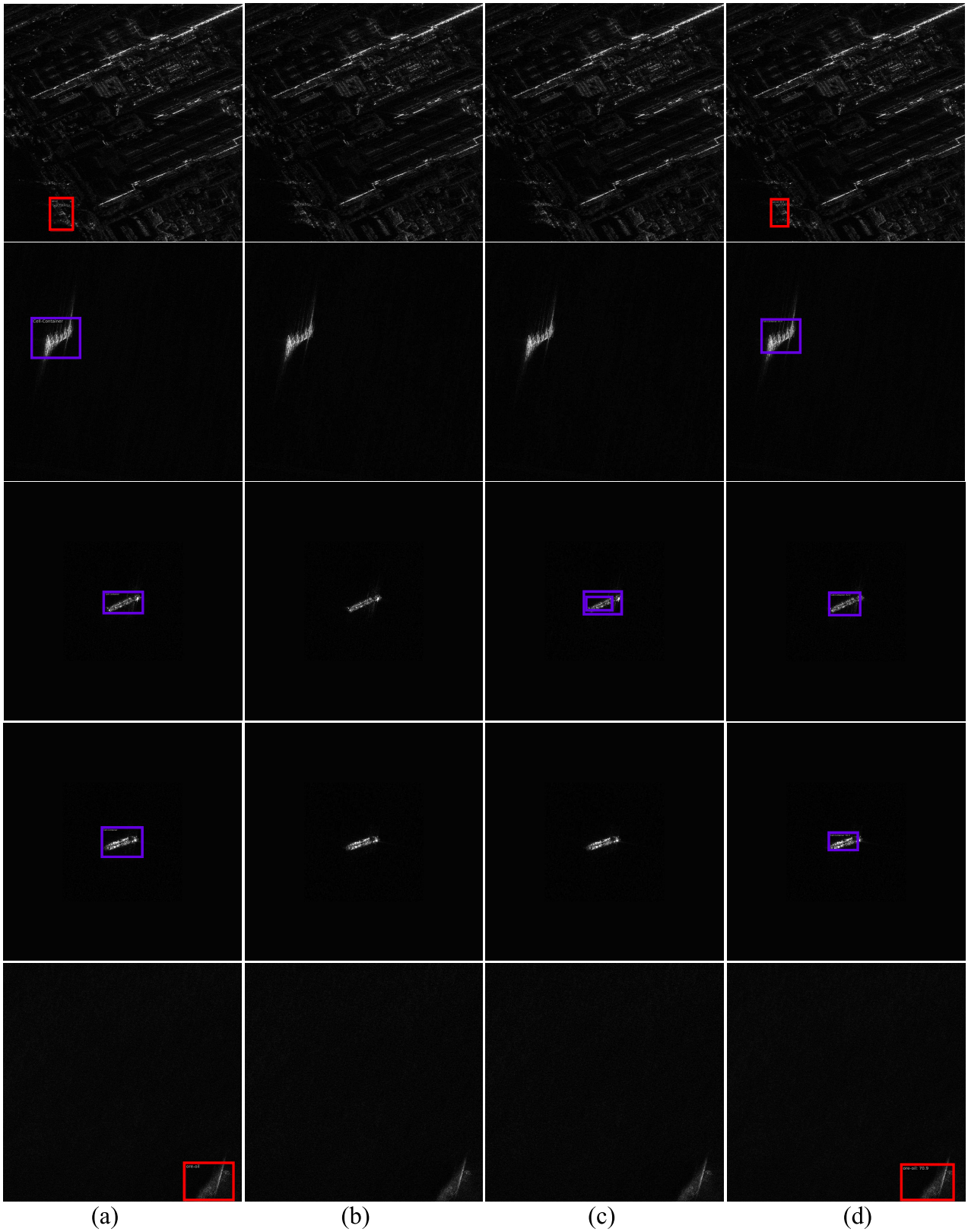


Fig. 7. ZSD results on own dataset. (a) Ground truth. (b) Result of BLC. (c) Results of ZSD-SCR. (d) Result of ours. All these objects belong to unseen classes.

semi-supervised learning, the network successfully transfers from the seen to the unseen domain, generating proposal boxes. In the second image, due to the high level of feature interference in the nearshore scene, fewer high-confidence

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

proposal boxes were generated, which may also affect subsequent detection results.

After evaluating Recall@100, we also evaluated the mAP for seen and unseen classes, as shown in Table V. The results align with those of the Recall@100 metric, indicating that POM achieves notable improvement in the unseen classes. The mAP for the seen classes increases from 55.21% to 66.2%, while the average mAP for the unseen classes rises from 6.64% to 21.59%. In general, POM combined with SELM maintains the proposal box transfer capability after nonmaximum suppression, significantly enhancing the detection performance of SAR-ZSD.

C. Comparison With Conventional ZSD Methods

To validate the effectiveness of the proposed method, we compared the SSRP network with several state-of-the-art methods in the field of optical ZSD, ranging from classic to the latest, i.e., BLC [65], ZSI [66], WEA-DINO [67], SUZOD [68], ZSD-SCR [69], and RRFS [12]. The quantitative results listed in Table VI suggest that SSRP achieves a better tradeoff between recall and mAP in the SAR-ZSD task, benefiting from the lightweight and effective module design. In particular, BLC, WEA-DINO, and ZSI aim to learn a mapping function that connects the visual features of seen classes with their corresponding semantic vectors, which is similar to our method. Their performance was not satisfactory, especially for BLC, which achieved the lowest recall@100 and mAP. SUZOD, ZSD-SCR, and RRFS, samples of unseen classes are created based on their semantics by a GAN [27] trained with seen class data, allowing conventional object detectors to identify unseen objects effectively. Our framework employs the SELM with adaptive attention weighting to extract SAR-specific features, followed by cross-modal visual–textual interaction. This design effectively enhances feature alignment between modalities, achieving superior performance over existing semantic embedding methods (BLC, ZSI, and WEA-DINO), as evidenced by significant improvements in mAP. The results of RRFS were second only to the proposed method. However, its generator needs to adversarially generate new images, which may result in higher memory and computational costs compared to most other methods using the same training and evaluation protocols. The SSRP method achieved the best results among all methods. From the perspective of cross-domain generation of detection boxes, SSRP's recall values stabilized around 67%. From the perspective of ZSD accuracy, SSRP improved by 19.59% compared to BLC. We can observe that low mAP is a common issue for all compared methods. This is because the object targets in remote sensing SAR images are very small and can easily be confused with complex backgrounds. Therefore, the functions of POM and SELM within the network, specifically their abilities to enhance cross-domain mapping and emphasize scattering features, are particularly advantageous for SAR-ZSD.

As shown in Table VI, our method significantly outperforms similar approaches that align visual and semantic features in terms of Recall@100 and mAP. This advantage arises from the weak response of traditional embedding-based methods to unseen classes in the SAR image domain. Our proposed POM enhances the recall rate for unseen classes. Although the recall results of the SUZOD, ZSD-SCR, and RRFS that used GAN to generate unseen classes are close to ours, with SUZOD performing the best at an IoU of 0.4, our method still achieves significantly higher mAP values, outperforming theirs by 9.79%, 6.89%, and 4.49%. In general, the POM used semi-supervised learning to enhance the recall of ZSL by leveraging unlabeled samples to enrich the feature space and improve sensitivity toward unseen categories. However, due to inherent ambiguities in visual–semantic alignment and substantial modality discrepancies between SAR imagery and textual descriptions, the accuracy of zero-shot recognition still has considerable room for improvement. Our future work will focus on designing such an embedding framework to improve both precision and recall in SAR-ZSD.

Fig. 7 shows a few examples of images on SRSDD that demonstrate the capability of our method to detect unseen objects in SAR scenarios. The leftmost column shows the ground truth, while the rightmost column shows the results from our method. Other methods struggled to generate proposal boxes for unseen classes during the feature perception stage, leading to failure in ZSD tasks. The first ore-oil image, situated in a nearshore scene, experienced the greatest scattering interference from man-made structures, making it particularly challenging to distinguish the foreground from the background. As a result, the confidence level of the detection boxes obtained by our method is slightly lower than for other images of the same class. For the cell-container class, which has relatively clean backgrounds, the ZSD-SCR method produced unstable results during the proposal stage due to the lack of scattering-focused learning. In contrast, our method, incorporating semi-supervised learning and scattering mechanisms, demonstrated accurate detection results across all images of unseen classes.

V. Conclusion

In this article, we present a scattering-attention semi-supervised-guided reinforced region proposal (SSRP) network for zero-shot object detection in SAR remote sensing imagery to alleviate the weak cross-domain feature response and scattering noise interference problems. First, the POM improves the capability of feature cross-domain migration by improving the response of the proposal box with the semi-supervised method. Second, the scatter excitation learning module is proposed to focus on the local scattering center features of the foreground examples to avoid the influence of background noise interference. In addition, we developed an engineering-level image-to-text method to generate descriptive terms for key attributes of SAR images, which is used to automatically expand the zero-shot training dataset. Meanwhile, we also improved the existing SRSDD dataset to promote the development of SAR-ZSD. The comprehensive experimental results confirm that the SSRP network outperforms the state-of-the-art generic methods and can achieve better cross-domain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

migration and detection results. In future work, we will explore the application of CLIP in the SAR domain.

REFERENCES

[1] X. Zhang, G. Gao, and S.-W. Chen, "Polarimetric autocorrelation matrix: A new tool for joint characterizing of target polarization and Doppler scattering mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5213522.

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 39, 2016, pp. 1137–1149.

[3] T. Zhang, J. Ji, X. Li, W. Yu, and H. Xiong, "Ship detection from PolSAR imagery using the complete polarimetric covariance difference matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2824–2839, May 2019.

[4] Z. Tian, W. Wang, K. Zhou, X. Song, Y. Shen, and S. Liu, "Weighted pseudo-labels and bounding boxes for semisupervised SAR target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5193–5203, 2024.

[5] G. Gao, Q. Bai, C. Zhang, L. Zhang, and L. Yao, "Dualistic cascade convolutional neural network dedicated to fully PolSAR image ship detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 663–681, Aug. 2023.

[6] G. Gao, C. Zhang, L. Zhang, and D. Duan, "Scattering characteristic-aware fully polarized SAR ship detection network based on a four-component decomposition model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5222722.

[7] C. Zhang, G. Gao, J. Liu, and D. Duan, "Oriented ship detection based on soft thresholding and context information in SAR images of complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5200615.

[8] A. Bansal, K. Sikka, G. Sharma, R. Chellappa, and A. Divakaran, "Zero-shot object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 384–400.

[9] Z. Li, L. Yao, X. Zhang, X. Wang, S. Kanhere, and H. Zhang, "Zero-shot object detection with textual descriptions," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, Jul. 2019, pp. 8690–8697.

[10] P. Zhu, H. Wang, and V. Saligrama, "Dont even look once: Synthesizing features for zero-shot detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11693–11702.

[11] Y. Li, P. Li, H. Cui, and D. Wang, "Inference fusion with associative semantics for unseen object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 3, pp. 1993–2001.

[12] P. Huang, J. Han, D. Cheng, and D. Zhang, "Robust region feature synthesizer for zero-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7622–7631.

[13] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[14] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[15] J. Deng, W. Wang, H. Zhang, T. Zhang, and J. Zhang, "PolSAR ship detection based on superpixel-level contrast enhancement," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[16] L. Chen, R. Luo, J. Xing, Z. Li, Z. Yuan, and X. Cai, "Geospatial transformer is what you need for aircraft detection in SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5225715.

[17] B. Huang, T. Zhang, S. Quan, W. Wang, W. Guo, and Z. Zhang, "Scattering enhancement and feature fusion network for aircraft detection in SAR images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1936–1950, Feb. 2025.

[18] X. Xu, T. Zhang, H. Liu, W. Guo, and Z. Zhang, "An information-expanding network for water body extraction based on U-Net," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[19] G. Gao, L. Yao, W. Li, L. Zhang, and M. Zhang, "Onboard information fusion for multisatellite collaborative observation: Summary, challenges, and perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 2, pp. 40–59, Jun. 2023.

[20] B. Shen, T. Liu, G. Gao, H. Chen, and J. Yang, "A low-cost polarimetric radar system based on mechanical rotation and its signal processing," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 2, pp. 4744–4765, Apr. 2025.

[21] S. Lei, D. Lu, X. Qiu, and C. Ding, "SRSDD-v1.0: A high-resolution SAR rotation ship detection dataset," *Remote Sens.*, vol. 13, no. 24, p. 5104, Dec. 2021.

[22] D. He, W. Guo, T. Zhang, Z. Zhang, and W. Yu, "Occluded target recognition in SAR imagery with scattering excitation learning and channel dropout," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[23] M. Elhoseiny, Y. Zhu, H. Zhang, and A. Elgammal, "Link the head to the 'beak': Zero shot learning from noisy text description at part precision," 2017, *arXiv:1709.01148*.

[24] C. Cao, L. Bao, G. Gao, G. Liu, and X. Zhang, "A novel method for ocean wave spectra retrieval using deep learning from Sentinel-1 wave mode data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4204016.

[25] G. Gao et al., "A multibranch embedding network with bi-classifier for few-shot ship classification of SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5201515.

[26] W. Xu, Y. Xian, J. Wang, B. Schiele, and Z. Akata, "VGSE: Visually-grounded semantic embeddings for zero-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9306–9315.

[27] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 214–223.

[28] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[29] Z. Han, Z. Fu, S. Chen, and J. Yang, "Contrastive embedding for generalized zero-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2371–2381.

[30] Z. Huang, X. Zhang, Z. Tang, F. Xu, M. Datcu, and J. Han, "Generative artificial intelligence meets synthetic aperture radar: A survey," *IEEE Geosci. Remote Sens. Mag.*, early access, Nov. 30, 2025, doi: 10.1109/MGRS.2024.3483459.

[31] J. Li, M. Jing, K. Lu, L. Zhu, Y. Yang, and Z. Huang, "Alleviating feature confusion for generative zero-shot learning," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1587–1595.

[32] X. Li et al., "VS-boost: Boosting visual-semantic association for generalized zero-shot learning," in *Proc. Thirty-Second Int. Joint Conf. Artif. Intell.*, Aug. 2023, pp. 1107–1115.

[33] G. Gao et al., "FDC-TA-DSN ship classification model and dataset construction based on complex-valued SAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7034–7047, 2025.

[34] G. Gao et al., "An oriented ship detection method of remote sensing image with contextual global attention mechanism and lightweight task-specific context decoupling," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4200918.

[35] Y. Guan et al., "A new indicator for assessing fishing ecological pressure using multi-source data: A case study of the South China sea," *Ecol. Indicators*, vol. 170, Jan. 2025, Art. no. 113096.

[36] C. Zhang et al., "Development and application of ship detection and classification datasets: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 4, pp. 12–45, Dec. 2024.

[37] G. Gao, Y. Dai, X. Zhang, D. Duan, and F. Guo, "ADCG: A cross-modality domain transfer learning method for synthetic aperture radar in ship automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5109114.

[38] H. Yang, S. Guo, X. Liu, Z. Yu, and Z. Huang, "Efficient SAR automatic target recognition based on self-distillation," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Nov. 2024, pp. 1–4.

[39] S. Wen, W. Zhao, F. Ji, R. Peng, L. Zhang, and Q. Wang, "Recognizing unknown disaster scenes with knowledge graph-based zero-shot learning (KG-ZSL) model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5621315.

[40] C. Yan, Q. Zheng, X. Chang, M. Luo, C.-H. Yeh, and A. G. Hauptman, "Semantics-preserving graph propagation for zero-shot object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 8163–8176, 2020.

[41] L. Liu, T. Zhou, G. Long, J. Jiang, and C. Zhang, "Attribute propagation network for graph zero-shot learning," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 4, pp. 4868–4875.

[42] G.-S. Xie et al., "Region graph embedding network for zero-shot learning," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 562–580.

[43] J. Guo, S. Guo, Q. Zhou, Z. Liu, X. Lu, and F. Huo, "Graph knows unknowns: Reformulate zero-shot learning as sample-level graph recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 6, pp. 7775–7783.

[44] X. Chen et al., "Explanatory object part aggregation for zero-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 2, pp. 851–868, Feb. 2024.

[45] Z. Huang, C. Wu, X. Yao, Z. Zhao, X. Huang, and J. Han, "Physics inspired hybrid attention for SAR target recognition," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 164–174, Jan. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.

[46] F. Pourpanah et al., "A review of generalized zero-shot learning methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4051–4070, Apr. 2023.

[47] L. Zhang et al., "Towards effective deep embedding for zero-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 9, pp. 2843–2852, Sep. 2020.

[48] Y. Xian, Z. Akata, G. Sharma, Q. Nguyen, M. Hein, and B. Schiele, "Latent embeddings for zero-shot classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 69–77.

[49] Z. Akata, S. Reed, D. Walter, H. Lee, and B. Schiele, "Evaluation of output embeddings for fine-grained image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2927–2936.

[50] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Fine-grained object recognition and zero-shot learning in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 770–779, Feb. 2018.

[51] Y. Li, Z. Zhu, J.-G. Yu, and Y. Zhang, "Learning deep cross-modal embedding networks for zero-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10590–10603, Dec. 2021.

[52] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[53] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2023, Art. no. 5642123.

[54] D. Huynh and E. Elhamifar, "Fine-grained generalized zero-shot learning via dense attribute-based attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4483–4493.

[55] Z. Liu, S. Guo, J. Guo, Y. Xu, and F. Huo, "Towards unbiased multi-label zero-shot learning with pyramid and semantic attention," *IEEE Trans. Multimedia*, vol. 25, pp. 7441–7455, 2023.

[56] G.-S. Xie, Z. Zhang, H. Xiong, L. Shao, and X. Li, "Towards zero-shot learning: A brief review and an attention-based embedding network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1181–1197, Mar. 2023.

[57] Q. Song and F. Xu, "Zero-shot learning of SAR target feature space with deep generative neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2245–2249, Dec. 2017.

[58] Q. Song, H. Chen, F. Xu, and T. J. Cui, "EM simulation-aided zero-shot learning for SAR automatic target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1092–1096, Jun. 2020.

[59] Q.-R. Wei, H. He, Y. Zhao, and J.-A. Li, "Learn to recognize unknown SAR targets from reflection similarity," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4002205.

[60] Q.-R. Wei, C.-Y. Chen, M. He, and H.-M. He, "Zero-shot SAR target recognition based on classification assistance," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[61] T. D. Ross, S. W. Worrell, V. J. Velten, J. C. Mossing, and M. L. Bryant, "Standard SAR ATR evaluation experiments using the MSTAR public release data set," *Proc. SPIE*, vol. 3370, pp. 566–573, Sep. 1998.

[62] M. M. Derakhshani et al., "Assisted excitation of activations: A learning technique to improve object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9193–9202.

[63] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[64] S. Rahman, S. Khan, and F. Porikli, "Zero-shot object detection: Learning to simultaneously recognize and localize novel concepts," in *Proc. Asian Conf. Comput. Vis.*, 2019, pp. 547–563.

[65] Y. Zheng, R. Huang, C. Han, X. Huang, and L. Cui, "Background learnable cascade for zero-shot object detection," in *Proc. Asian Conf. Comput. Vis.*, 2021, pp. 107–123.

[66] Y. Zheng, J. Wu, Y. Qin, F. Zhang, and L. Cui, "Zero-shot instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2593–2602.

[67] G. Wang, H. Zhao, Q. Chang, S. Lyu, G. Cheng, and H. Chen, "WEA-DINO: An improved DINO with word embedding alignment for remote scene zero-shot object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[68] N. Hayat, M. Hayat, S. Rahman, S. Khan, S. W. Zamir, and F. S. Khan, "Synthesizing the unseen for zero-shot object detection," in *Proc. Asian Conf. Comput. Vis.*, 2021, pp. 155–170.

[69] S. Sarma, S. Kumar, and A. Sur, "Resolving semantic confusions for improved zero-shot detection," in *Proc. 33rd Brit. Mach. Vis. Conf.*, Nov. 2022, pp. 231–236.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:13:26 UTC from IEEE Xplore. Restrictions apply.